

03 - 01. Flutter 開發環境 (一) - Windows 環境

一、準備開發 Flutter

目前 Flutter 的環境只有在 Mac OS 作業系統下才可以同時開發 iOS 與 Android 兩種平台的應用程式，在 Windows 作業系統上無法執行 iOS 應用程式的開發。

依據 Flutter 官網介紹在 Windows 作業系統上開發 Flutter 需安裝下列軟體或套件：

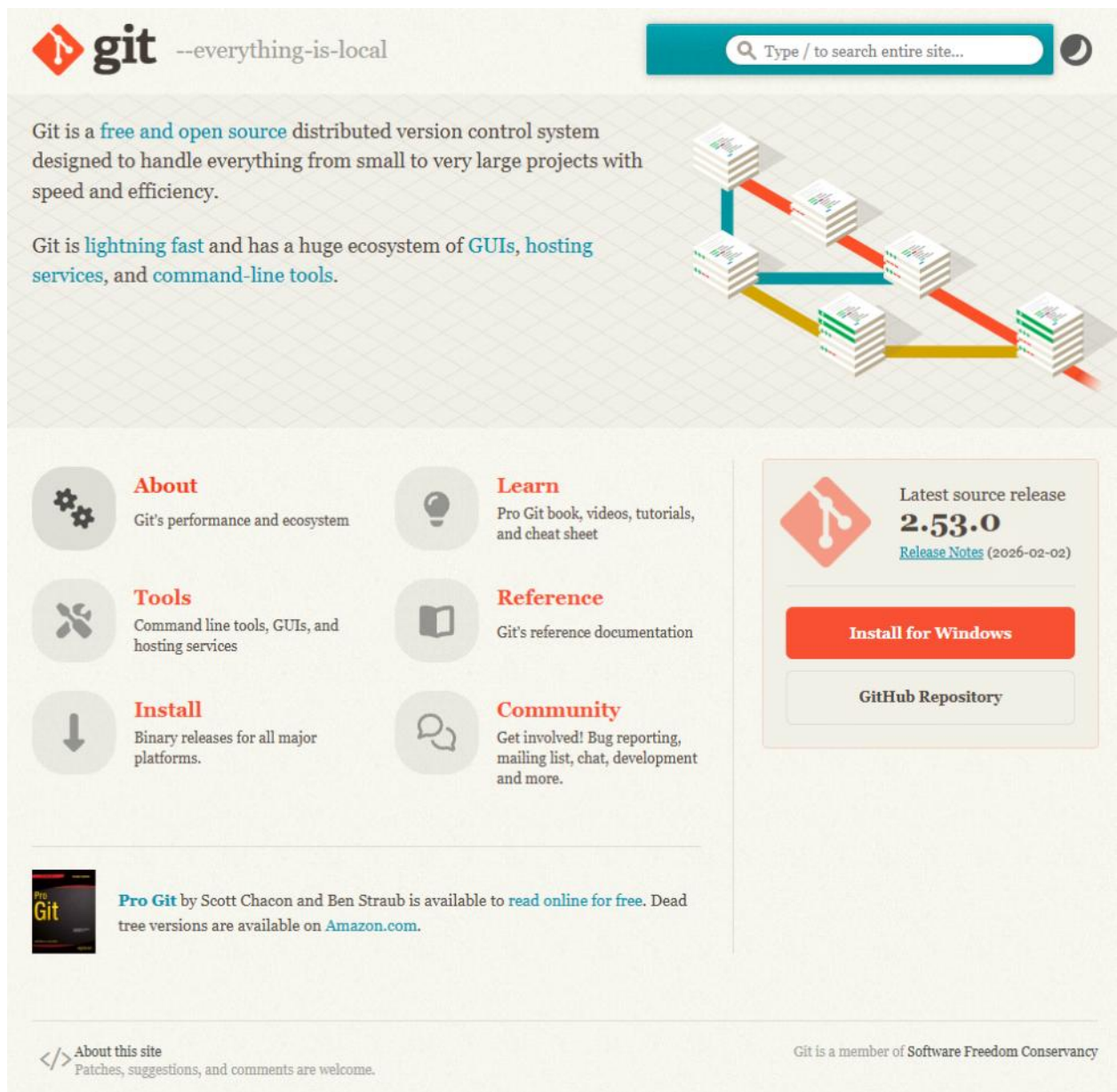
1. 開發環境：
 - Git
 - Flutter SDK
2. 開發軟體：
 - Visual Studio Code
3. 測試軟體：
 - Android Studio (僅使用其模擬器)



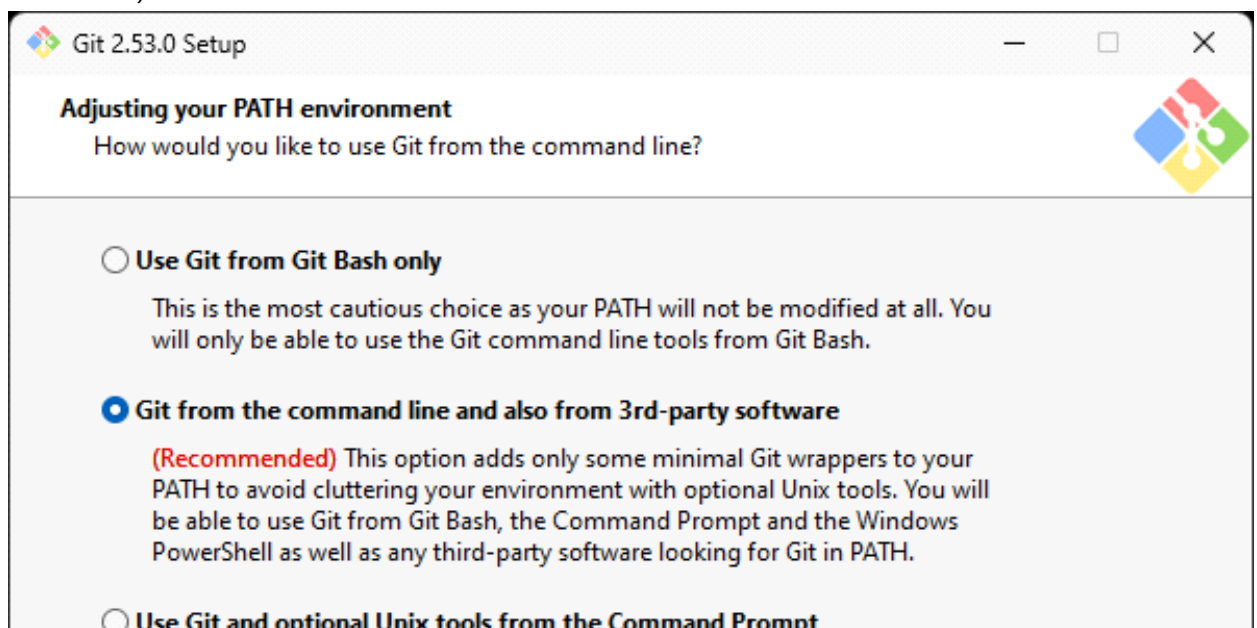
二、軟體安裝與設定

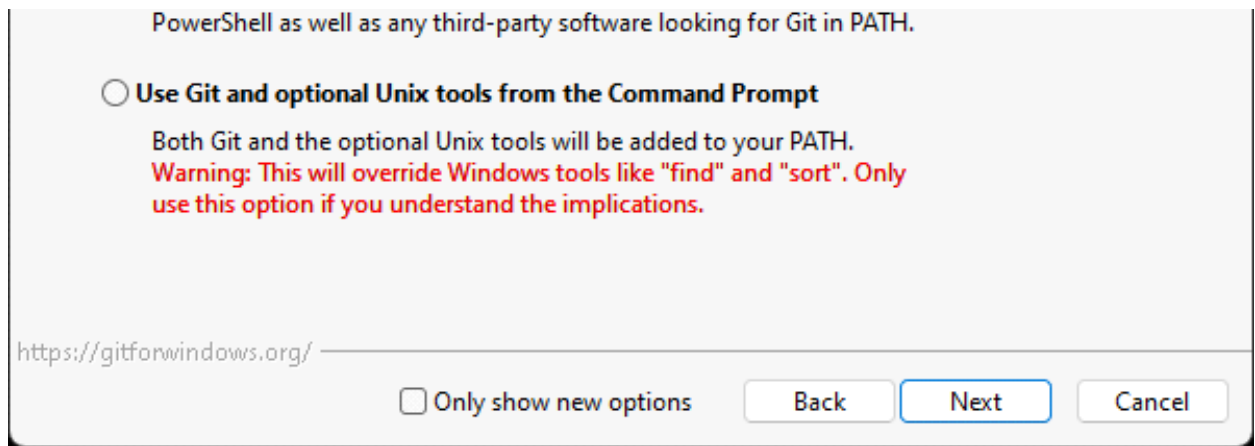
(一) Git

Git 是協助版本控制的軟體，協助下載與管理 Flutter 框架的原始碼，到 Git 的官網下載軟體。(<https://git-scm.com/>)。



下載後即可進行安裝，並於安裝過程選項中，在「Adjusting your PATH environment」這一步，請確保選擇了 "Git from the command line and also from 3rd-party software"（通常是預設選項），這樣才能讓 Flutter 順利呼叫 Git。





(二) Flutter SDK

1. 前往官網下載 SDK (<https://docs.flutter.dev/install/manual>) :

依據作業系統 (Windows/macOS) , 下載最新穩定版 (Stable channel) 的 Flutter SDK 壓縮檔。

Install and set up Flutter

To install the Flutter SDK, download the latest bundle from the SDK archive, then extract the SDK to where you want it stored.

1 Download the Flutter SDK bundle

Download the following installation bundle to get the latest stable release of the Flutter SDK.

[flutter_windows_3.41.4-stable.zip](#)

2. 解壓縮與放置：

將解壓縮後的 flutter 資料夾，放置到一個不需系統管理員權限的路徑，例如 D:\dev\flutter。切勿放在 C:\Program Files\ 這類需要高權限的目錄（若 C 槽空間不足，建議放置其他磁區）。

3. 設定環境變數 (至關重要！)

這一步是為了讓你的電腦終端機 (Terminal/CMD) 能辨識 flutter 這個指令。

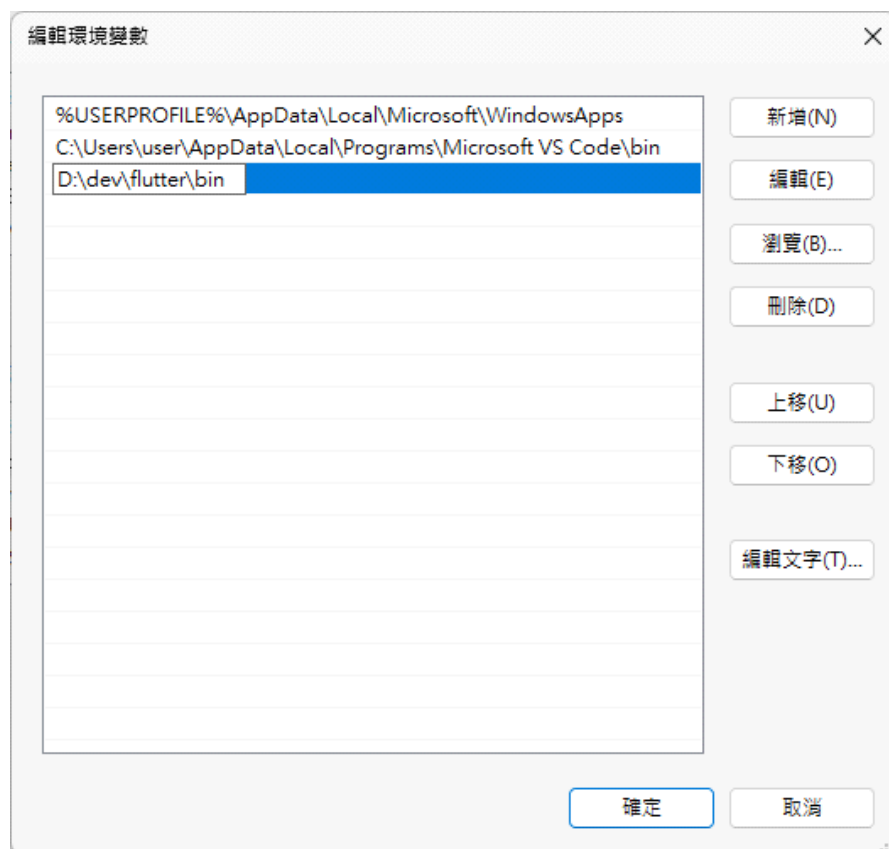
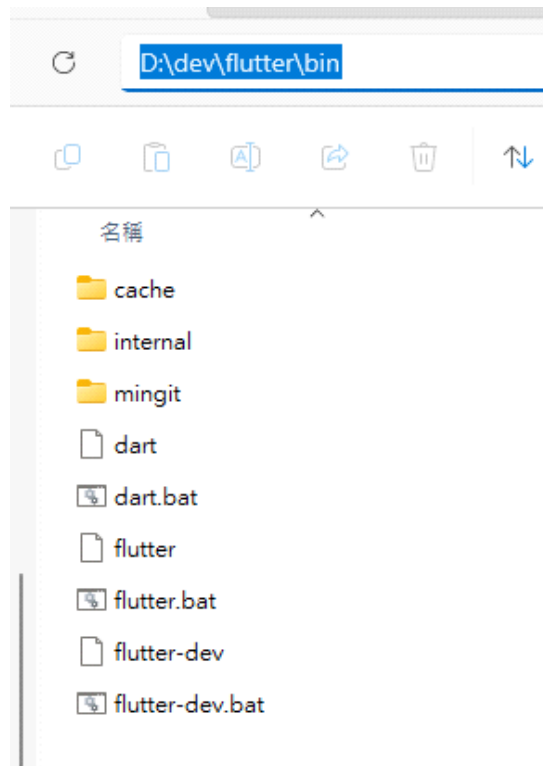
a. 開始功能表搜尋列輸入「env」或「環境變數」。

b. 選擇「編輯系統環境變數」。

c. 在彈出視窗中點擊「環境變數...」。

d. 在「使用者變數」區塊，找到 Path 變數，點擊「編輯」。

e. 點擊「新增」，並將你剛剛放置 Flutter 的路徑加上 \bin 貼上，例如：D:\dev\flutter。



4.開啟終端機，輸入 `flutter --version` 測試，如果能看到版本號（如下圖）就是成功了！

```
命令提示字元 - flutter --versi X + v
Microsoft Windows [版本 10.0.22621.4317]
(c) Microsoft Corporation. 著作權所有，並保留一切權利。

C:\Users\user>flutter --version
Building flutter tool...
Running pub upgrade...
Resolving dependencies...
Downloading packages... (12.5s)
Got dependencies.
Flutter 3.41.4 • channel stable • https://github.com/flutter/flutter.git
Framework • revision ff37bef603 (6 days ago) • 2026-03-03 16:03:22 -0800
Engine • hash 99578ad0355da00edb26301c874a3c250a5716f5 (revision e4b8dca3f1) (6 days ago) • 2026-03-03 18:24:54.000Z
Tools • Dart 3.11.1 • DevTools 2.54.1

C:\Users\user>
```

5. 神醫指令：flutter doctor

Flutter 提供了一個極其強大的診斷工具：flutter doctor。它會檢查環境並告知還缺少哪些東西。在終端機輸入後會看到類似下圖的畫面，其中 [✓] 代表正常，[X] 代表需要修復。

```
命令提示字元 - flutter --versi X + v
C:\Users\user>flutter doctor
Doctor summary (to see all details, run flutter doctor -v):
[✓] Flutter (Channel stable, 3.41.4, on Microsoft Windows [版本 10.0.22621.4317], locale zh-TW)
[✓] Windows Version (11 專業版 64 位元, 22H2, 2009)
[X] Android toolchain - develop for Android devices
    X Unable to locate Android SDK.
      Install Android Studio from: https://developer.android.com/studio/index.html
      On first launch it will assist you in installing the Android SDK components.
      (or visit https://flutter.dev/to/windows-android-setup for detailed instructions).
      If the Android SDK has been installed to a custom location, please use
      'flutter config --android-sdk' to update to that location.

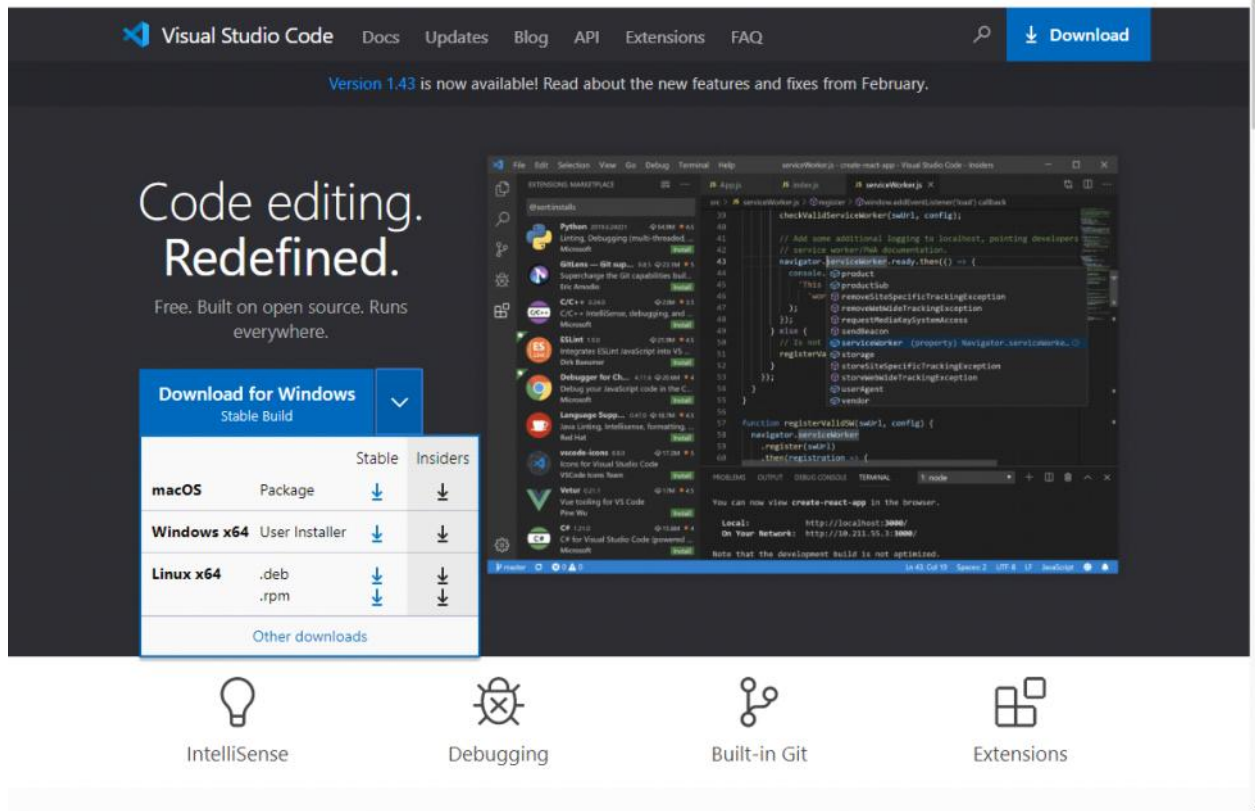
[✓] Chrome - develop for the web
[X] Visual Studio - develop Windows apps
    X Visual Studio not installed; this is necessary to develop Windows apps.
      Download at https://visualstudio.microsoft.com/downloads/.
      Please install the "Desktop development with C++" workload, including all of its
      default components
[✓] Connected device (3 available)
[✓] Network resources

! Doctor found issues in 2 categories.

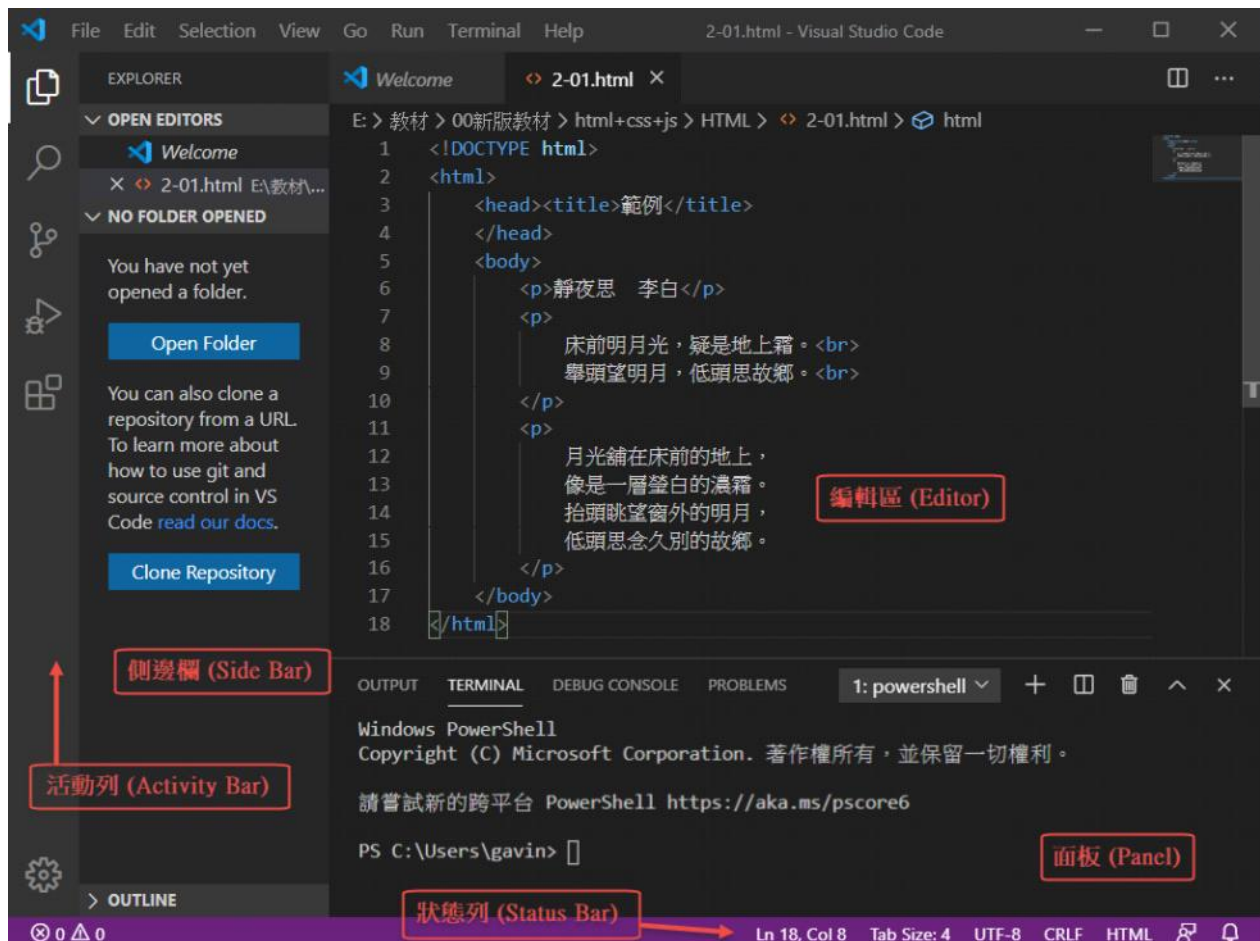
C:\Users\user>
```

(三) Visual Studio Code

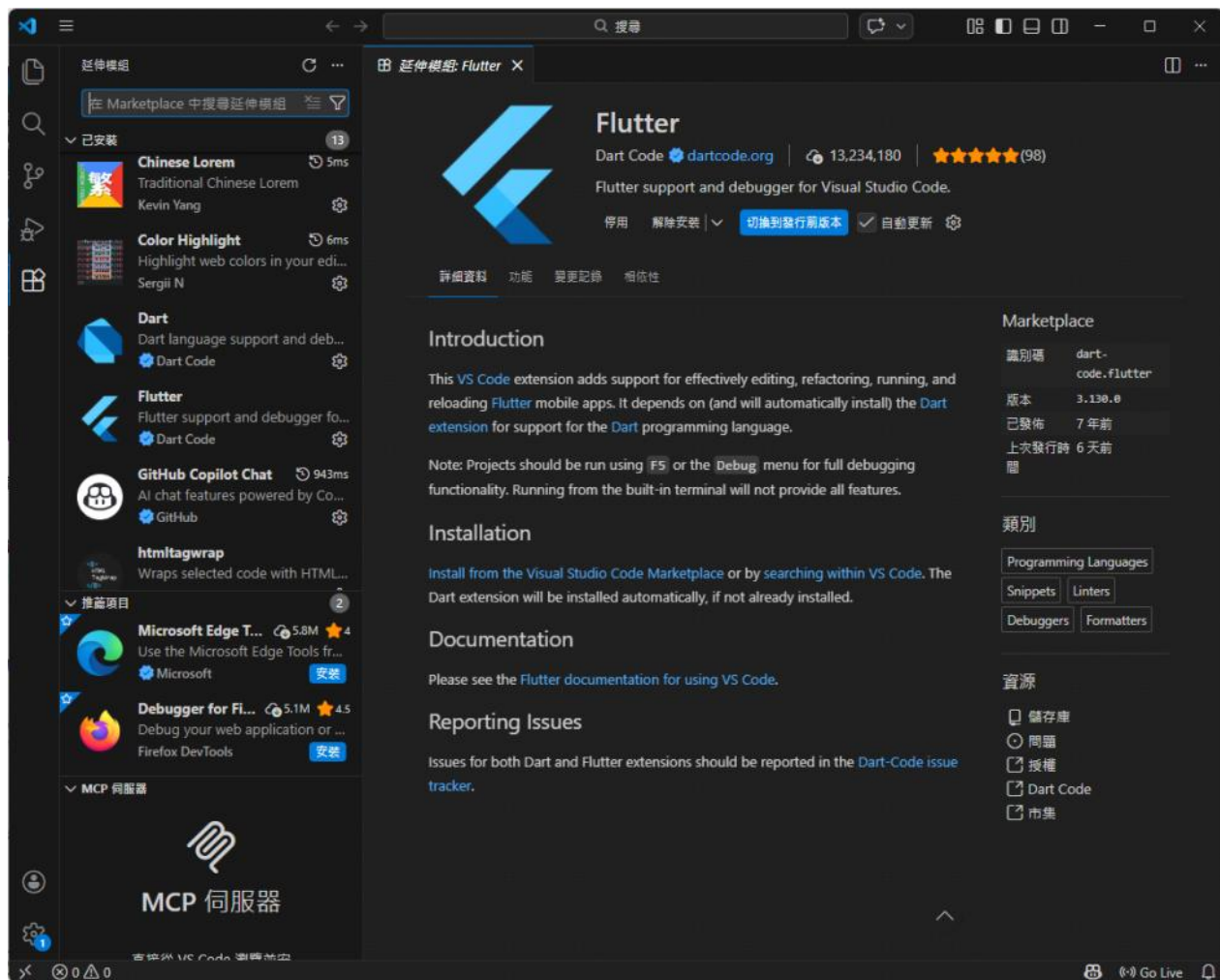
1. 進入 Visual Studio Code 網站 (<https://code.visualstudio.com>) 下載。



2. 下載後即可進行安裝，開啟畫面如下：



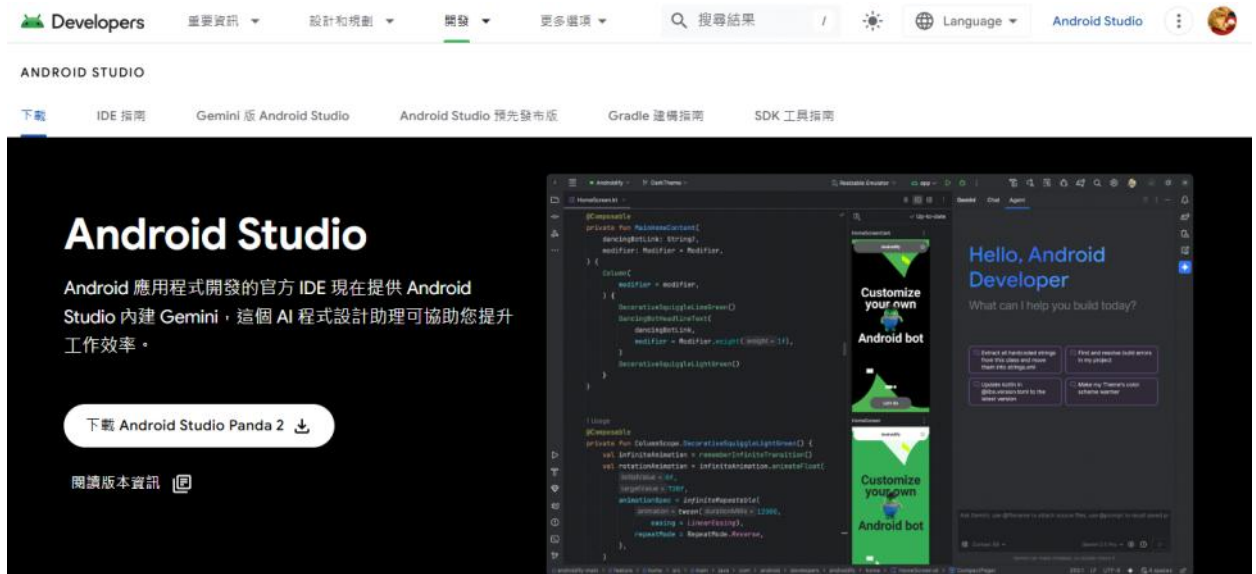
3.安裝 Flutter 外掛擴充套件，搜尋「Flutter」，選擇由 Dart Code 發布的 Flutter 擴充套件並安裝，它會自動一併安裝 Dart 擴充套件。



(四) 安裝 Android Studio 與 Android SDK

Android Studio 是一個具有圖形化界面的程式碼編輯開發平台，在 Android Studio 中可以編寫程式碼，也可以用它來進行專案的儲存、測試、除錯、甚至封裝成執行檔的工作，當然最重要的是其為免費，因此在 Android 官方網站中，建議使用 Android Studio 整合環境做為 Android 應用程式的開發平台，但目前只是利用它的模擬器進行測試。

1.Android Studio 官網 (<https://developer.android.com/studio?hl=zh-tw>) 下載並安裝。



新功能

功能

在 **Android Studio** 中試用 **Gemini**

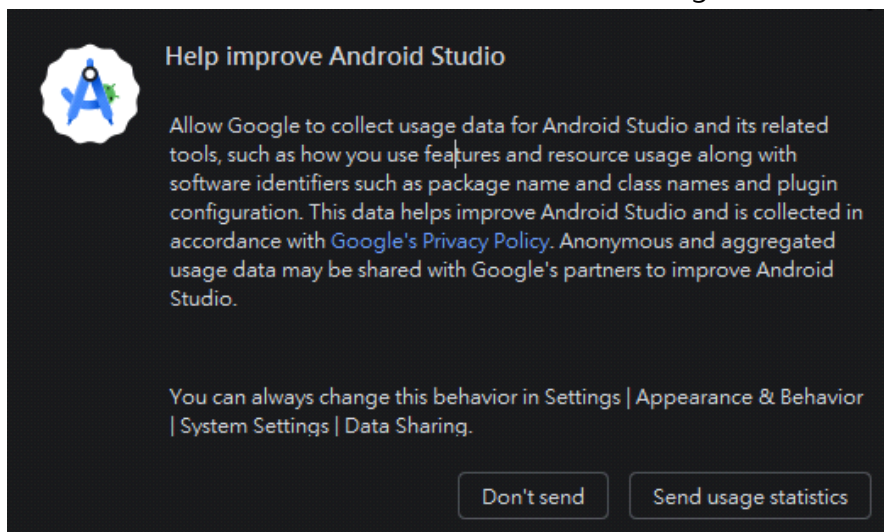
Gemini 版 Android Studio 是一個 AI 助理，協助您生成程式碼、修正程式碼，以

功能

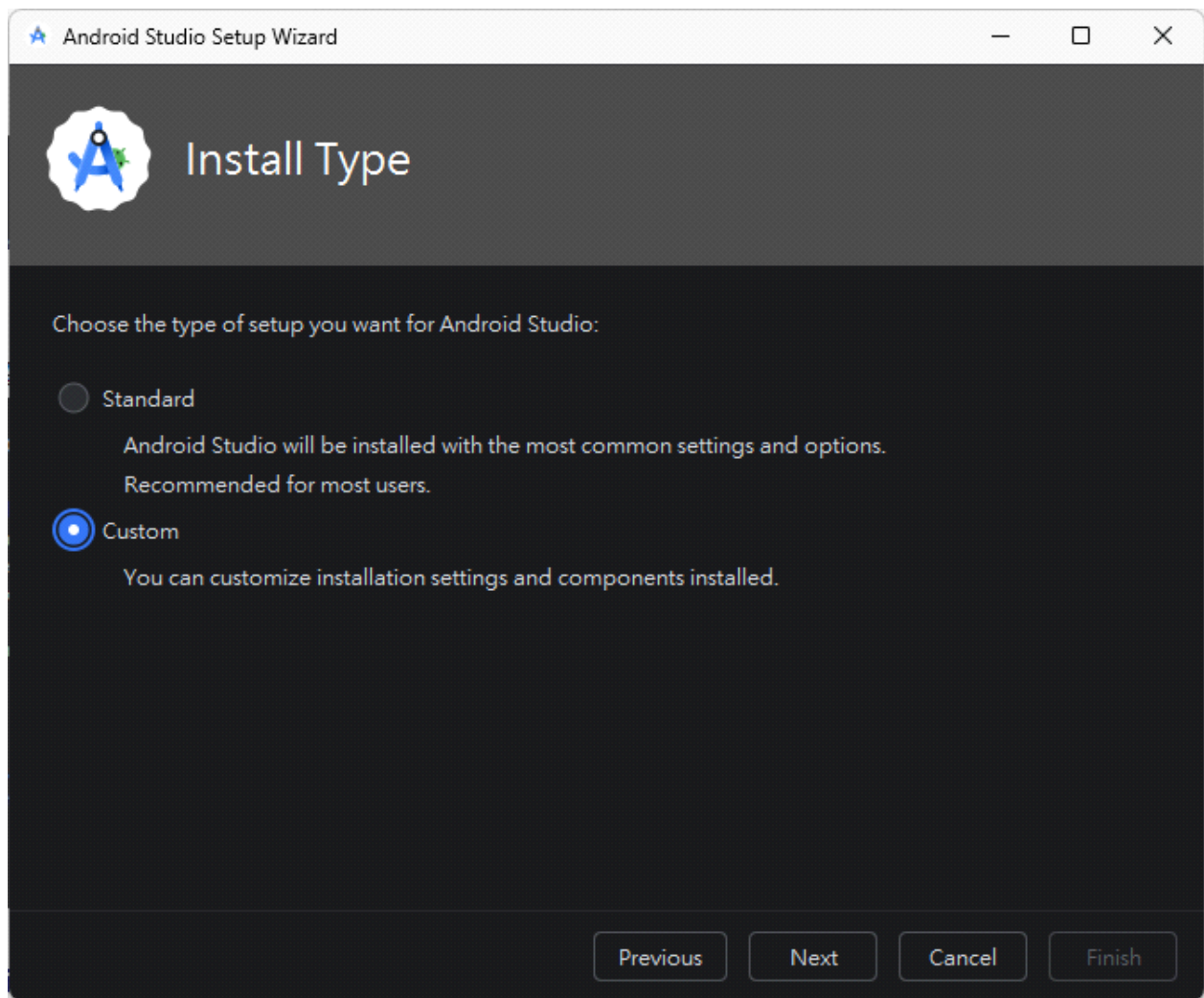
透過 **Android Studio Cloud** 隨時隨地編寫程式碼

您可以直接在瀏覽器中透過 **Firebase Studio** 試用 **Android Studio Cloud**，簡化

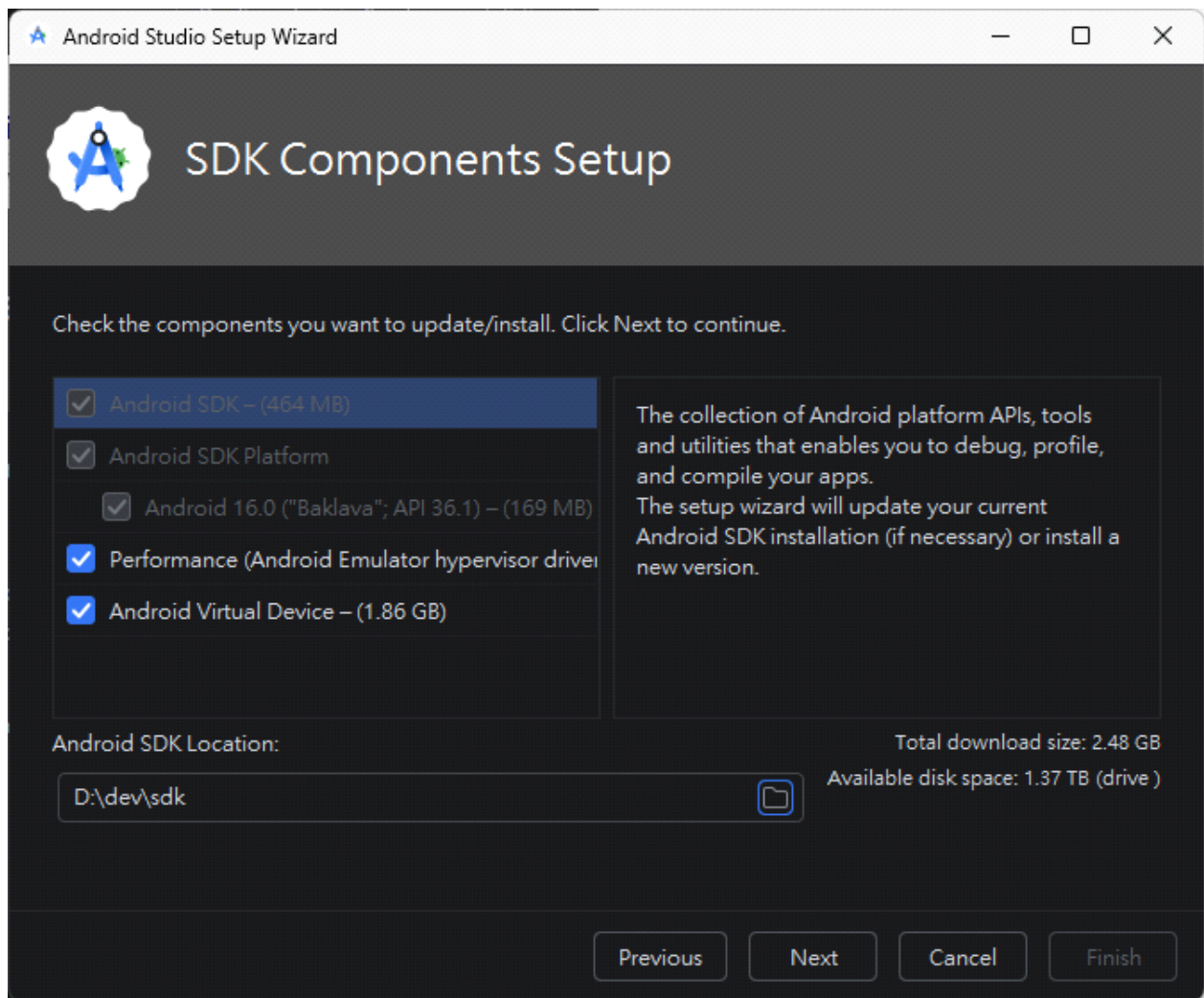
2.第一次啟動會進入設定畫面，首先是要不要與 Google 分享軟體使用資訊。



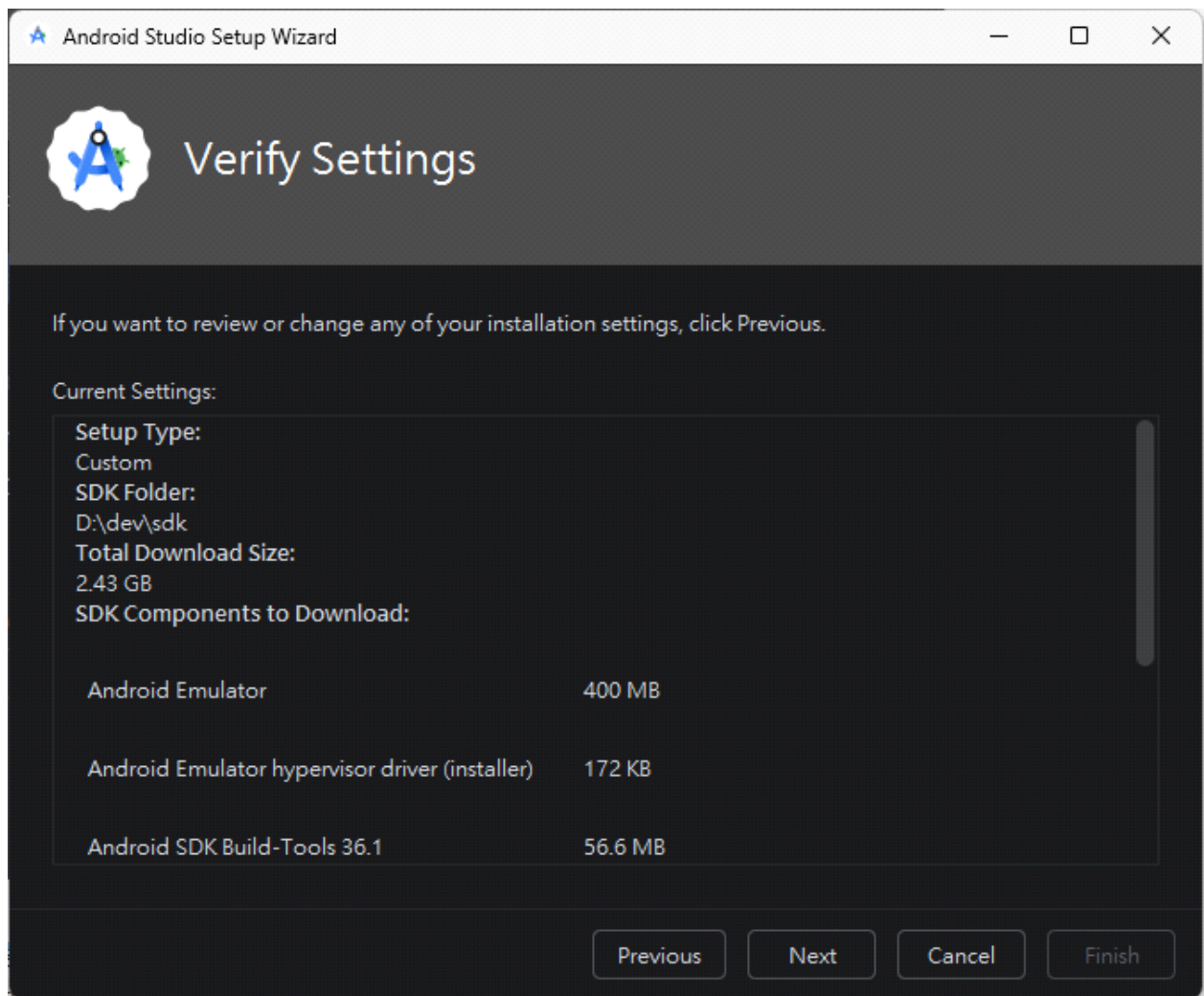
3.再來是安裝 Android SDK，選擇 Custom 後進入下一步。



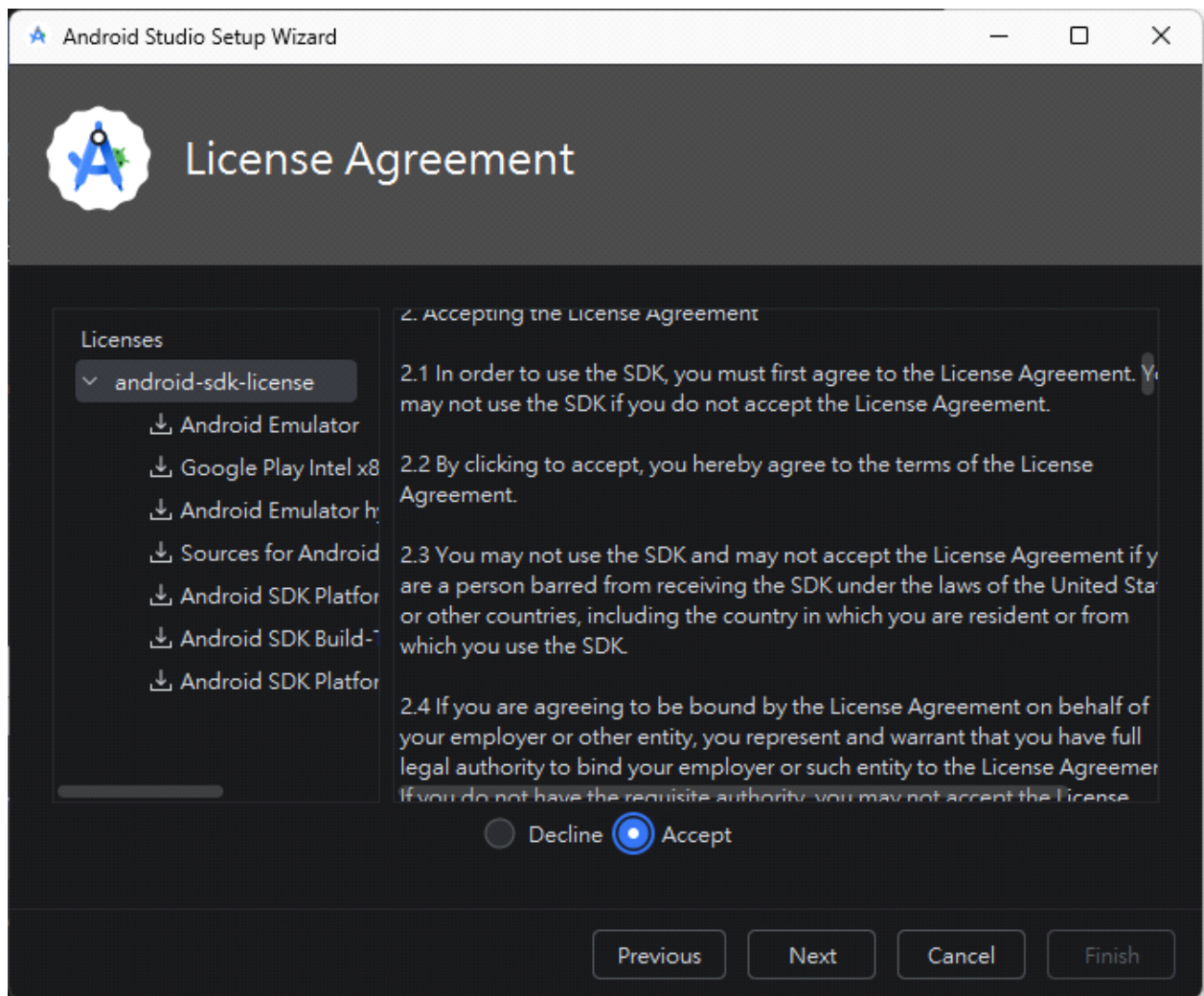
4.因 Android SDK 容量較大，若 C 槽空間不足，建議放置其他磁區。



5.確認安裝項目。

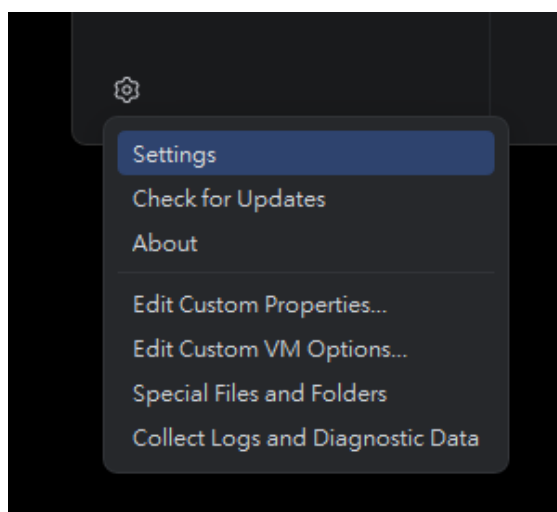


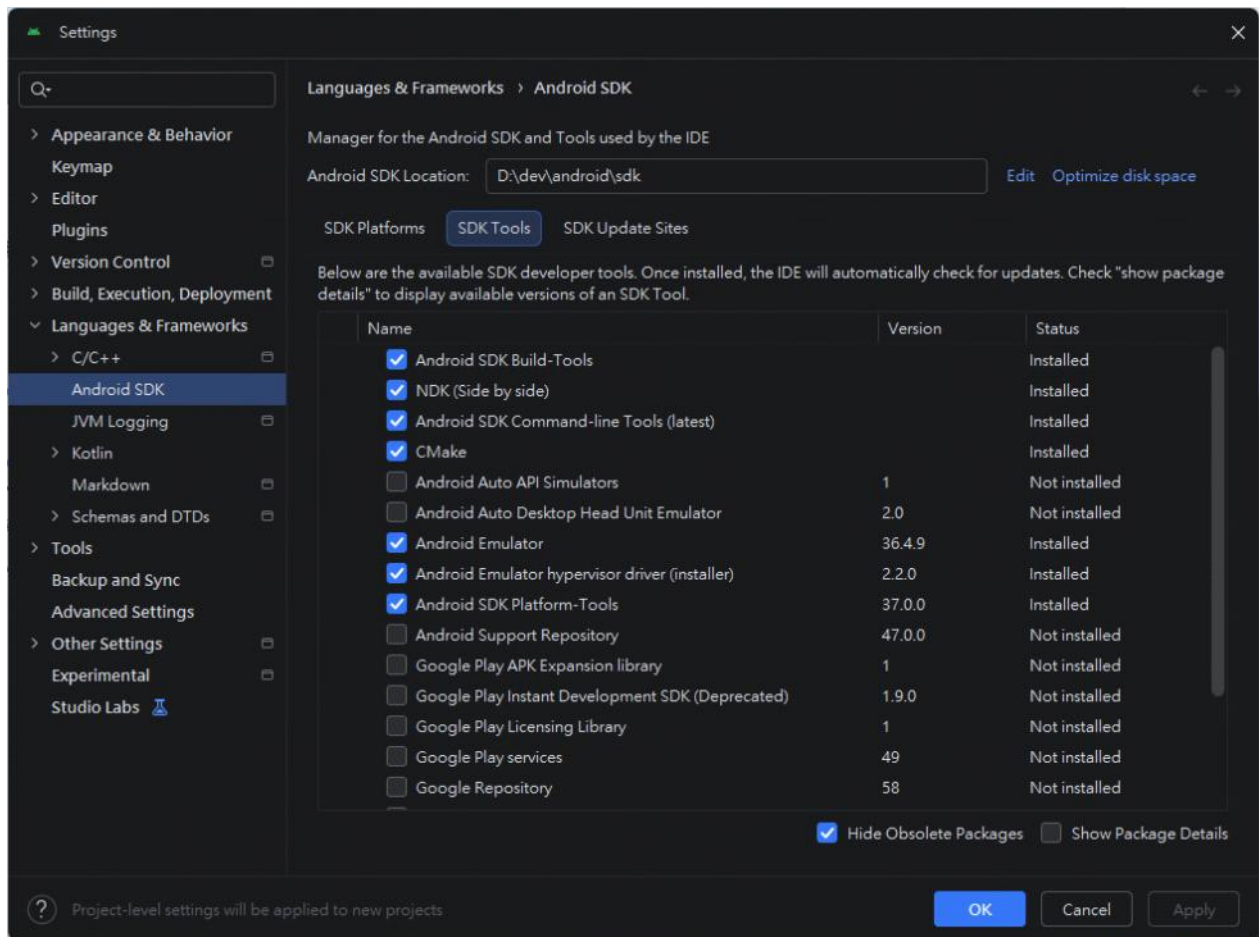
6.點選 Accept 後進行安裝



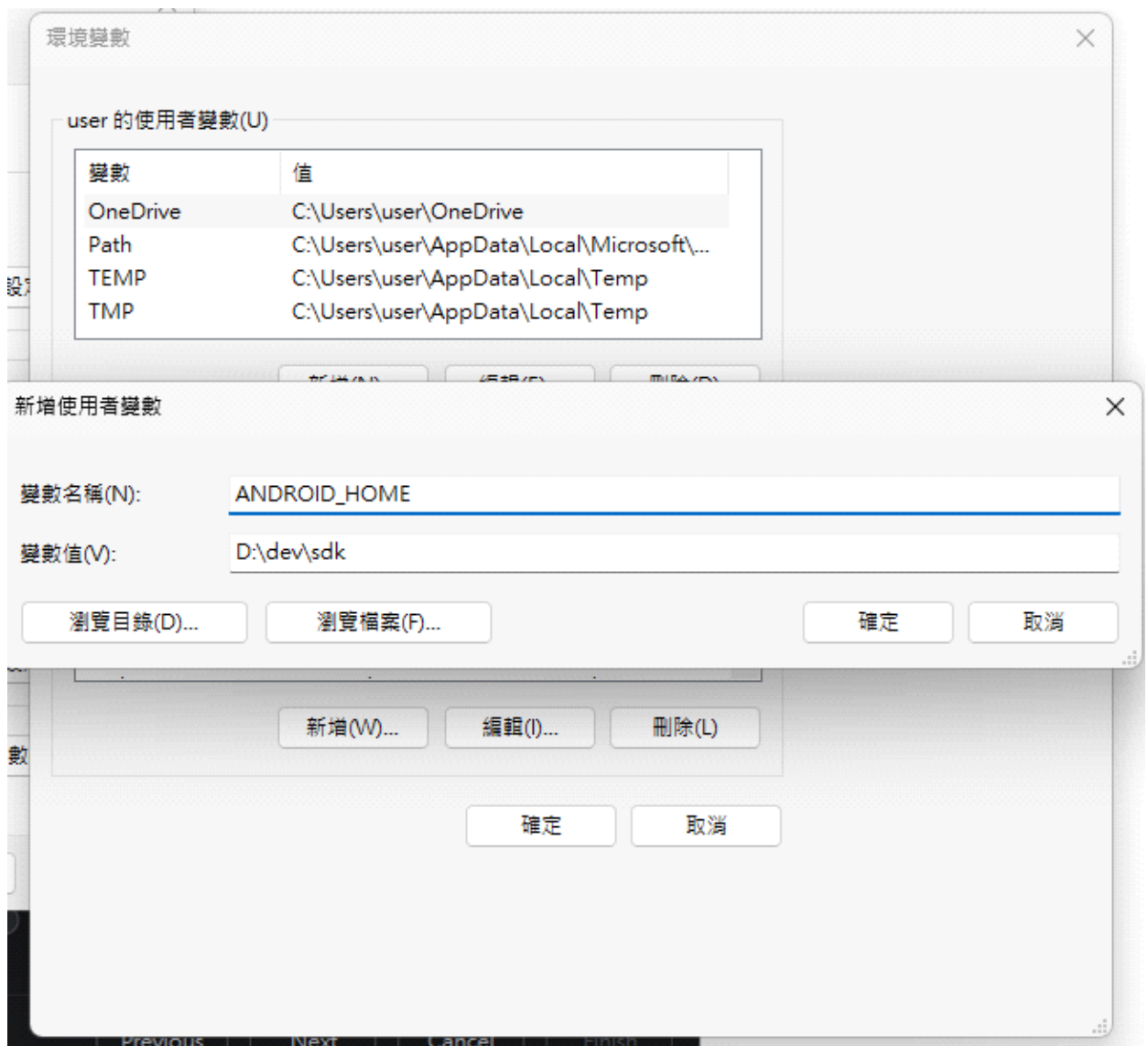
7.安裝 Android SDK 工具，點選設定後進入，輸入 sdk ，點選 SDK Tools，選擇以下項目安裝：

- Android SDK Build-Tools
- Android SDK Command-line Tools
- Android Emulator
- Android SDK Platform-Tools
- CMake
- NDK (Side by side)





8.新增 Android SDK 到環境變數。



9.執行授權指令：flutter doctor --android-licenses · 全部輸入 y 即可。

```

命令提示字元 - flutter doctor
ANY PRE-RELEASE MATERIALS ARE NON-QUALIFIED AND, AS SUCH, ARE PROVIDED AS IS AND AS AVAILABLE, POSSIBLY WITH FAULTS, AND WITHOUT REPRESENTATION OR WARRANTY OF ANY KIND.

10.8 Open Source Software. In the event Open Source software is included with Evaluation Software, such Open Source software is licensed pursuant to the applicable Open Source software license agreement identified in the Open Source software comments in the applicable source code file(s) and/or file header as indicated in the Evaluation Software. Additional detail may be available (where applicable) in the accompanying on-line documentation. With respect to the Open Source software, nothing in this Agreement limits any rights under, or grants rights that supersede, the terms of any applicable Open Source software license agreement.
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Accept? (y/N): y
All SDK package licenses accepted

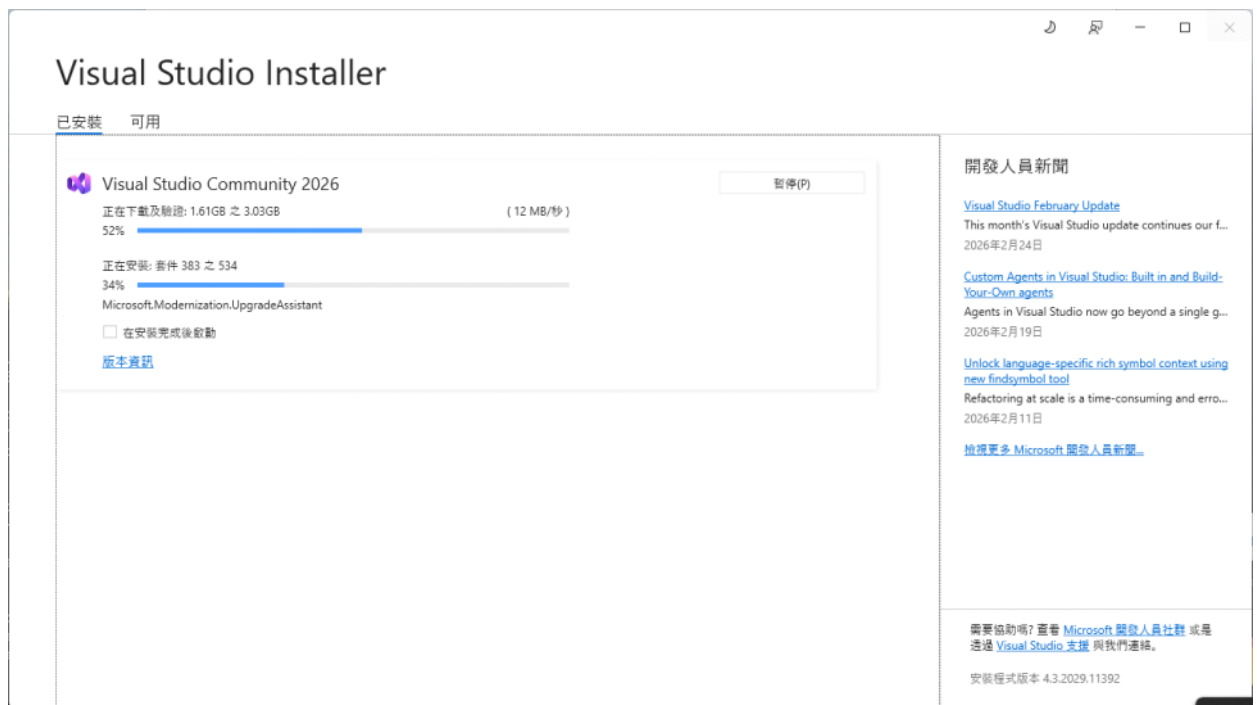
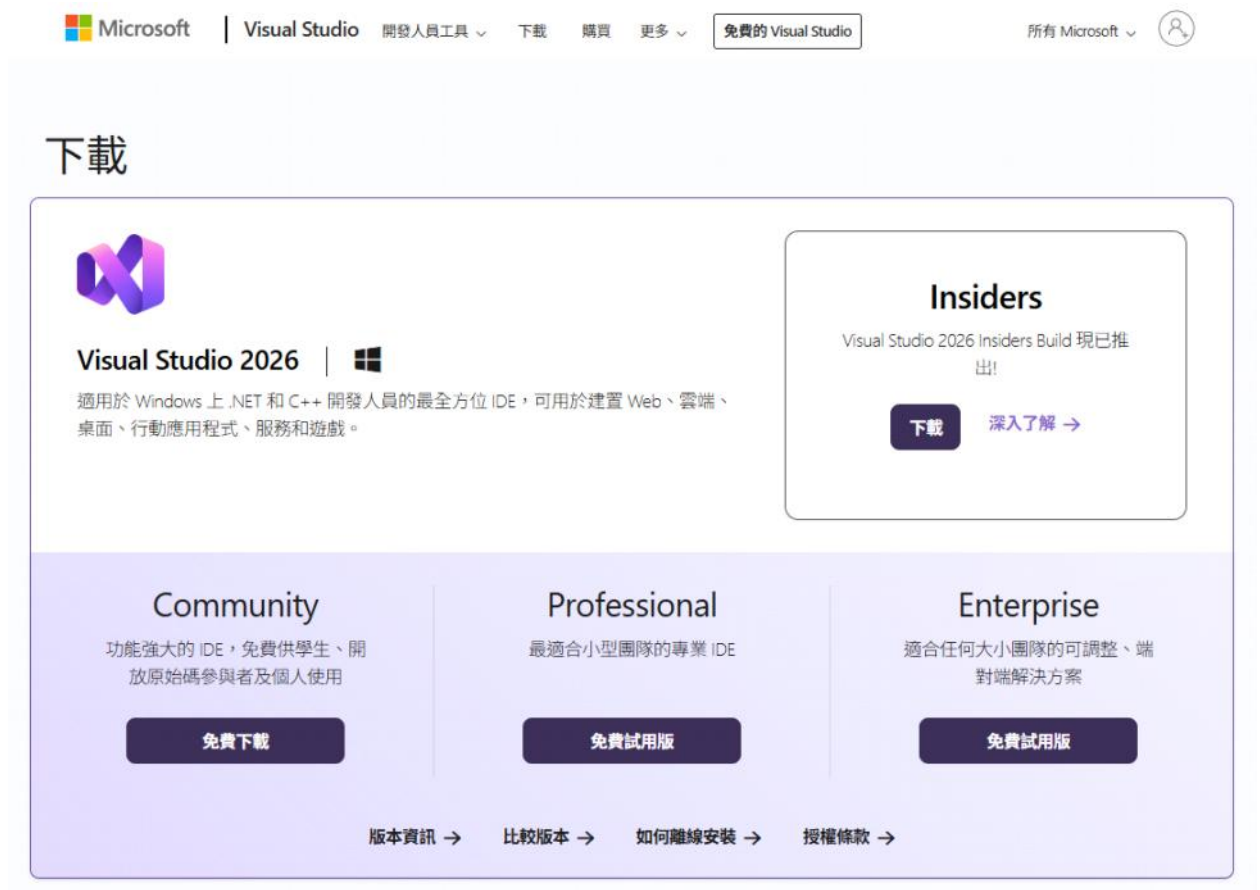
C:\Users\user>flutter doctor
Doctor summary (to see all details, run flutter doctor -v):
[✓] Flutter (Channel stable, 3.41.4, on Microsoft Windows [版本 10.0.26200.7840], locale zh-TW)
[✓] Windows Version (11 專業版 64 位元, 25H2, 2009)
[✓] Android toolchain - develop for Android devices (Android SDK version 36.1.0)
[✓] Chrome - develop for the web
[✓] Visual Studio - develop Windows apps
[X] Visual Studio not installed; this is necessary to develop Windows apps.
   Download at https://visualstudio.microsoft.com/downloads/.
   Please install the "Desktop development with C++" workload, including all of its default components
[✓] Connected device (3 available)
[✓] Network resources

! Doctor found issues in 1 category.

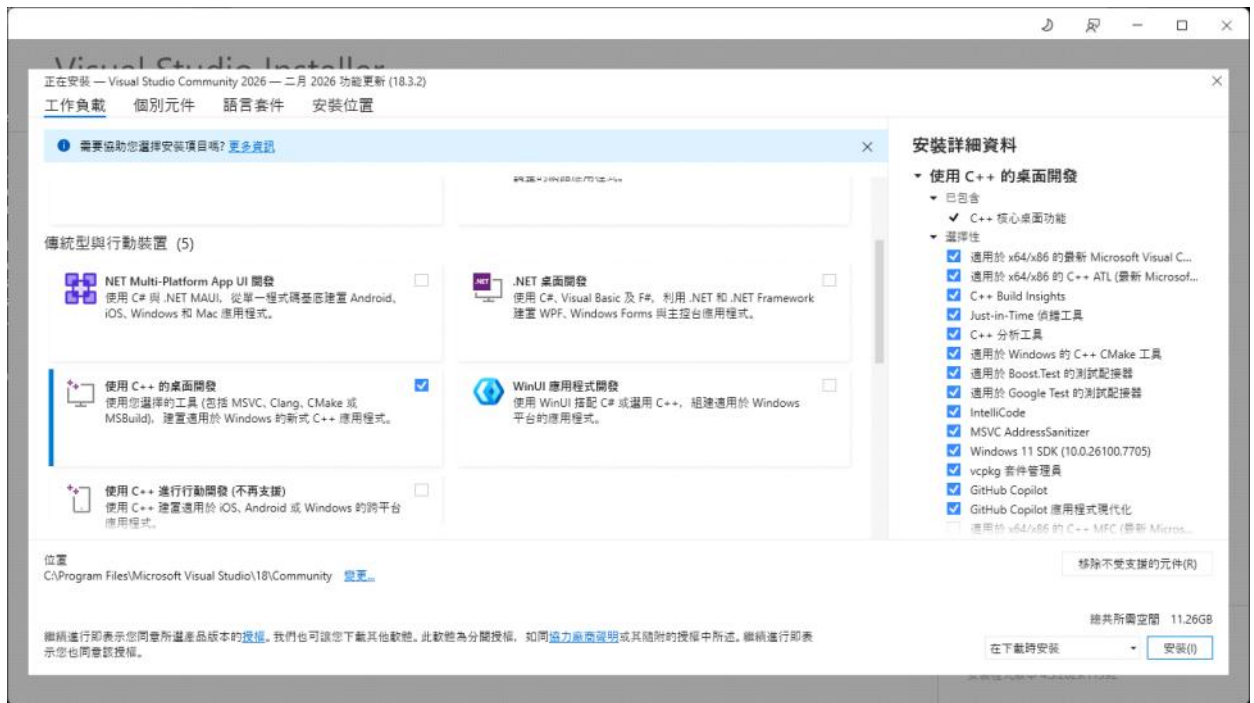
C:\Users\user>

```

10. 下載並安裝 Visual Studio (<https://visualstudio.microsoft.com/zh-hant/downloads/>) 。



在工作負載找到 使用 C++ 的桌面開發 (Desktop development with C++)，打勾之後按底下的安裝



11.執行flutter doctor · 確認全部選項都打勾

```

C:\Users\user>flutter doctor
Doctor summary (to see all details, run flutter doctor -v):
[✓] Flutter (Channel stable, 3.41.4, on Microsoft Windows [版本 10.0.26200.7840], locale zh-TW)
[✓] Windows Version (11 專業版 64 位元, 25H2, 2009)
[✓] Android toolchain - develop for Android devices (Android SDK version 36.1.0)
[✓] Chrome - develop for the web
[X] Visual Studio - develop Windows apps
    X Visual Studio not installed; this is necessary to develop Windows apps.
      Download at https://visualstudio.microsoft.com/downloads/.
      Please install the "Desktop development with C++" workload, including all of its default components
[✓] Connected device (3 available)
[✓] Network resources

! Doctor found issues in 1 category.

C:\Users\user>flutter doctor
Doctor summary (to see all details, run flutter doctor -v):
[✓] Flutter (Channel stable, 3.41.4, on Microsoft Windows [版本 10.0.26200.7840], locale zh-TW)
[✓] Windows Version (11 專業版 64 位元, 25H2, 2009)
[✓] Android toolchain - develop for Android devices (Android SDK version 36.1.0)
[✓] Chrome - develop for the web
[✓] Visual Studio - develop Windows apps (Visual Studio Community 2026 18.3.2)
[✓] Connected device (4 available)
[✓] Network resources

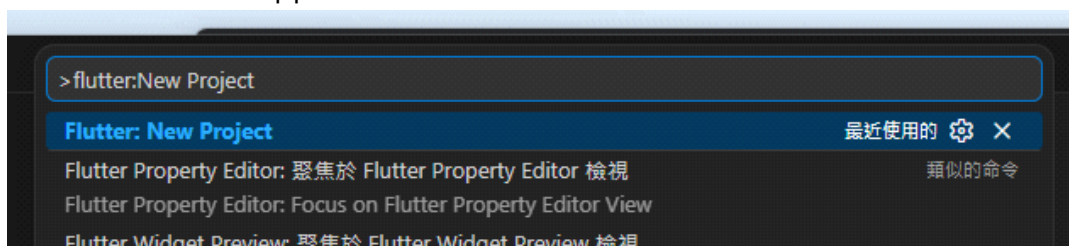
• No issues found!

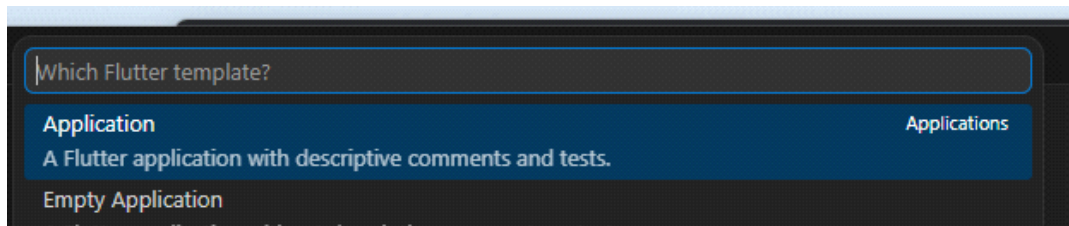
C:\Users\user>

```

三、範例專案執行

(一) 開啟 Visual Studio Code · 在命令列輸入 Flutter: New Project · VS Code 會提示「選擇哪個 Flutter 範本？」· 選擇 Application 範本即可快速建立一個簡單的計數器應用程式。





(二) 在終端機輸入 flutter run 啟動測試。

